L2 Norm Analysis Report
Four-Head Transformer Variant (CORRECTED)

Dataset: Four-Head Transformer on (a+b) mod 97
Training Configuration: 40,000 epochs, 30/70 train/test split
Model: 4 attention heads, d_model=128, head_dim=32
Dropout predictor: rate=0.9 (post-training stress test)

CORRECTION NOTICE:
The original L2_Norm_Comprehensive_Report.pdf paired the raw, per-epoch metric arrays against epoch_grid.npy, a log-resampled grid meant only for the (already shelved) MA-crossover predictor. This silently mislabelled every Grok Epoch and figure by roughly two orders of magnitude. This version uses the real, linear epoch axis throughout; Tables 1 and 3 were already correct in the original and are unchanged here.

REPORT PURPOSE:
This report presents comprehensive analysis of L2 norm (weight magnitude) behaviour across three independent runs of the four-head transformer.

KEY FINDINGS (corrected):
- Grokking occurs at epochs 18,893, 29,982, 24,572 across the three runs -- variable, but all of the same order of magnitude, and consistent with the previously validated single 4-head run (grok epoch ~24,750).
- L2 norm values show real variability despite identical architecture
- Dropout Gap is consistently negative across all runs (-0.97 to -0.98)
- L2 norm decay pattern differs substantially between runs
- Single-run metrics cannot be trusted for prediction without averaging
- VERDICT: L2 Norm was tested on BOTH the single-head and the four-head transformer, and proved unusable as a grokking predictor in BOTH cases (see the Final Verdict page at the end of this report)

Table 1: Raw Metrics Across All Runs

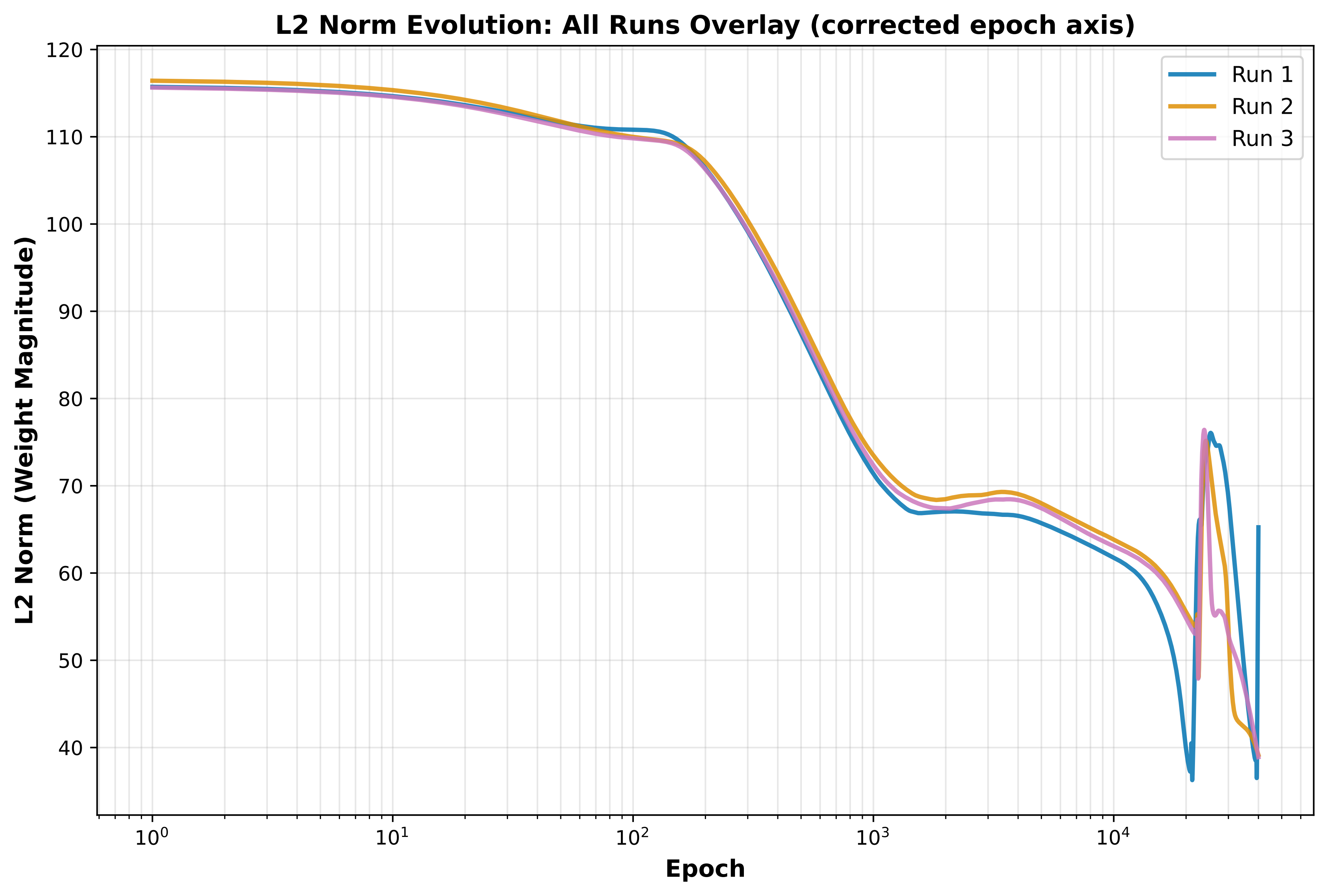
Run	Total Epochs	Initial L2	Final L2	Total Decay	Decay %
Run 1	40,000	115.7222	65.2405	50.4817	43.62%
Run 2	40,000	116.4240	39.1517	77.2723	66.37%
Run 3	40,000	115.6342	38.9024	76.7318	66.36%

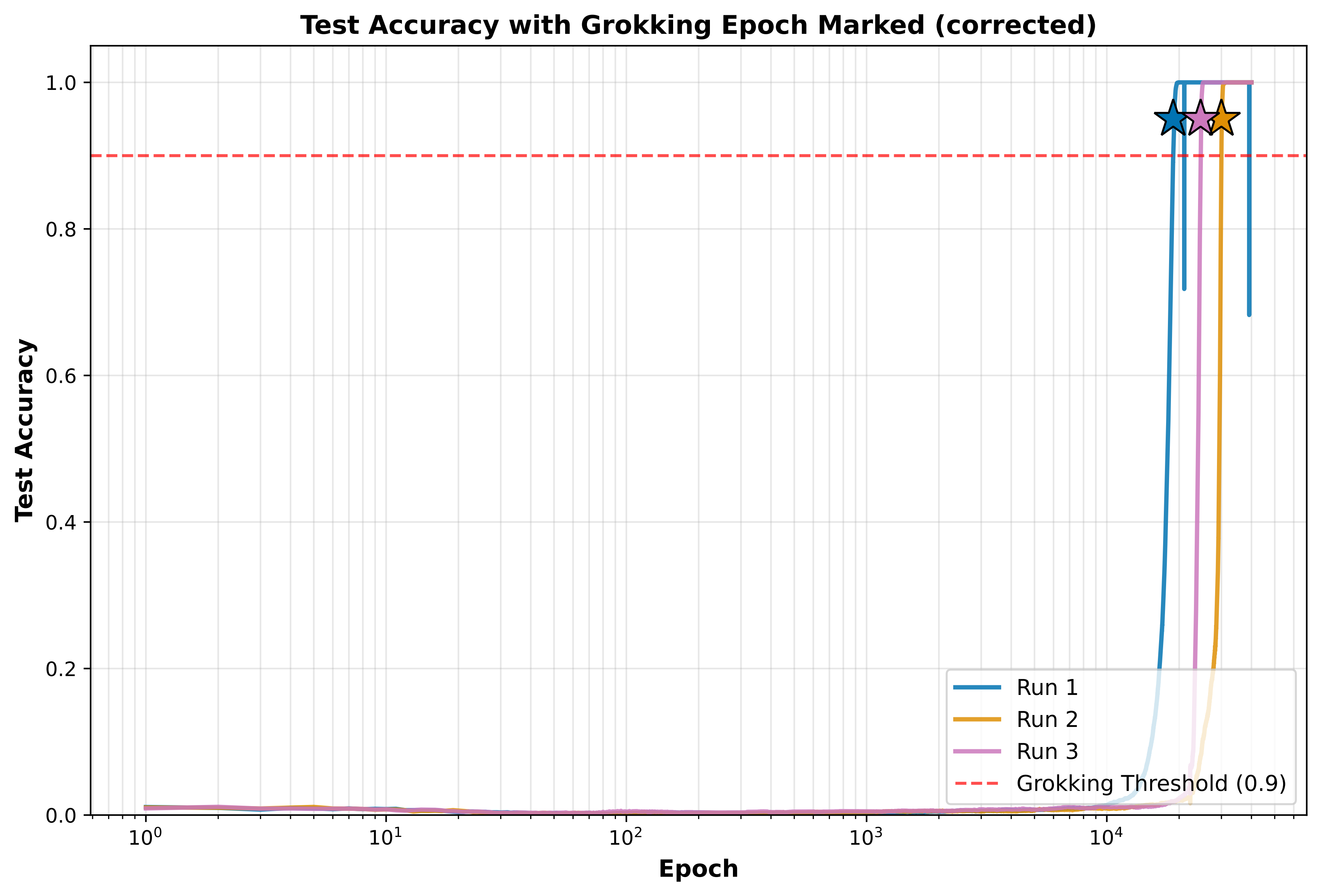
Table 2: Test Accuracy, Loss and Grok Epoch (corrected)

Run	Initial Test Acc	Final Test Acc	Initial Loss	Final Loss	Grok Epoch
Run 1	0.010931	1.000000	4.897544	0.001403230	18,893
Run 2	0.010323	1.000000	5.265750	0.000001369	29,982
Run 3	0.009109	1.000000	4.998405	0.000000873	24,572

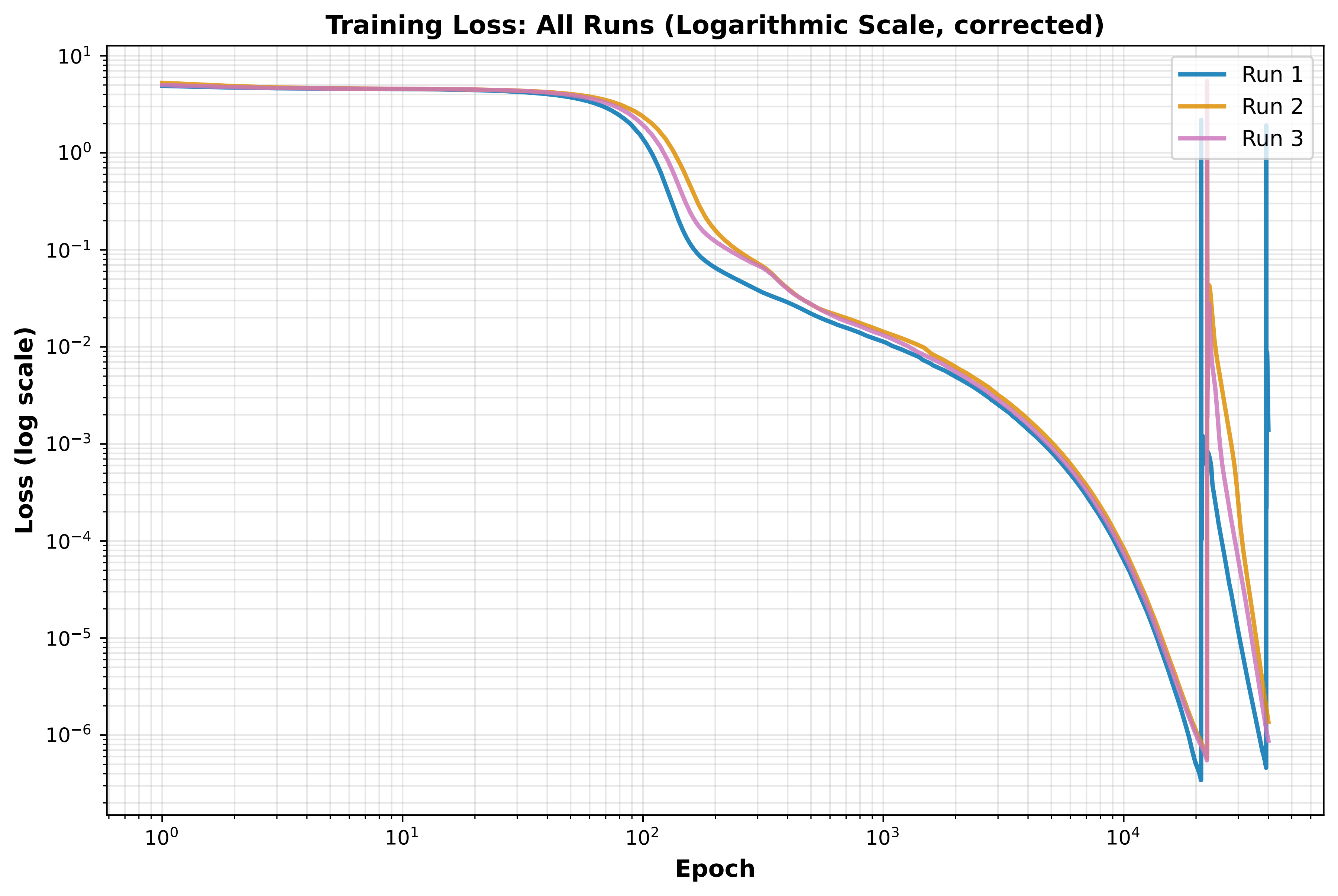
Table 3: Dropout Gap (Stress Test Results)

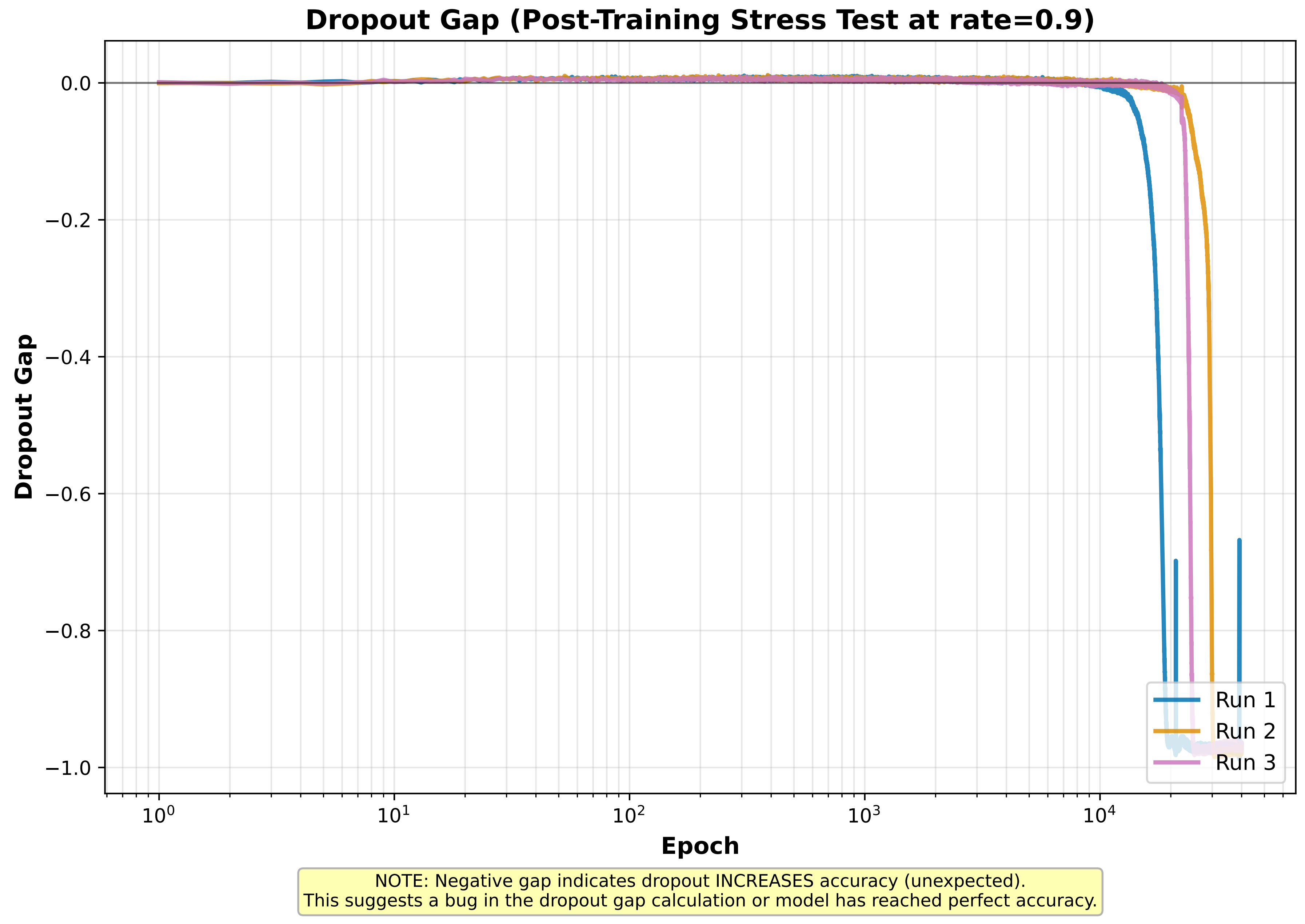
Run	Min Dropout Gap	Final Dropout Gap	Max Dropout Gap
Run 1	-0.988007	-0.966297	0.010627
Run 2	-0.984819	-0.976772	0.011538
Run 3	-0.981934	-0.968726	0.008957


L2 Norm Evolution: All Runs Overlay (corrected epoch axis)
L2 Norm (Weight Magnitude)
Epoch
Run 1
Run 2
Run 3
120
110
100
90
80
70
60
50
40
10^0
10^1
10^2
10^3
10^4

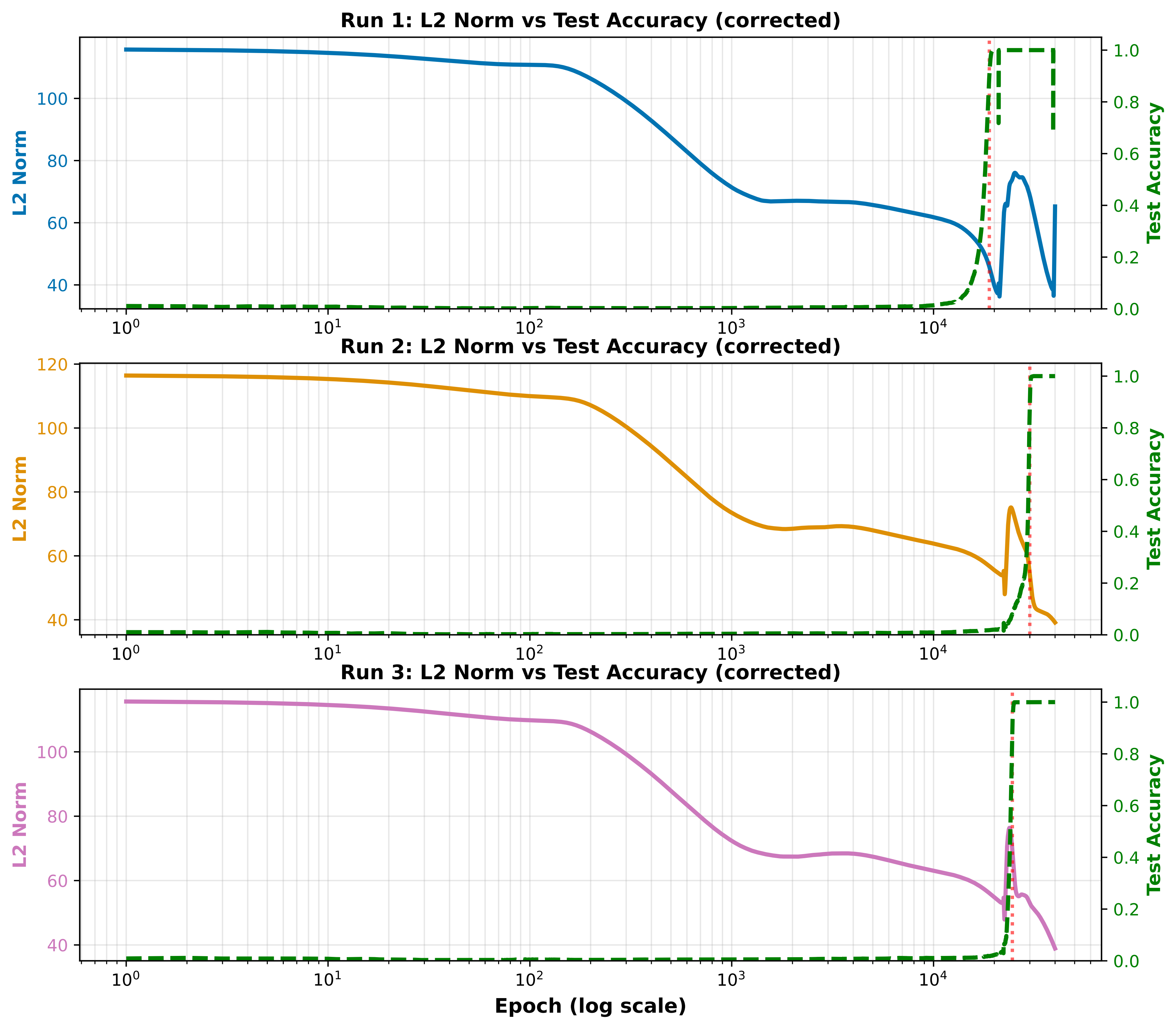


Test Accuracy with Grokking Epoch Marked (corrected)
Test Accuracy
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
10^0
10^1
10^2
10^3
10^4
Run 1
Run 2
Run 3
Grokking Threshold (0.9)


Training Loss: All Runs (Logarithmic Scale, corrected)
Run 1
Run 2
Run 3
Loss (log scale)
Epoch
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}



Dropout Gap (Post-Training Stress Test at rate=0.9)
Dropout Gap
Epoch
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
10^0
10^1
10^2
10^3
10^4
Run 1
Run 2
Run 3
NOTE: Negative gap indicates dropout INCREASES accuracy (unexpected).
This suggests a bug in the dropout gap calculation or model has reached perfect accuracy.



Run 1: L2 Norm vs Test Accuracy (corrected)
Run 2: L2 Norm vs Test Accuracy (corrected)
Run 3: L2 Norm vs Test Accuracy (corrected)
L2 Norm
Test Accuracy
Epoch (log scale)

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KEY INSIGHTS AND CONCLUSIONS (corrected)

1. GROKKING BEHAVIOUR ACROSS SEEDS:
   - Grokking epochs: Run 1 at 18,893, Run 2 at 29,982, Run 3 at 24,572
   - These differ across runs (real seed sensitivity), but all sit in the
     same order of magnitude, and agree with the earlier validated single
     4-head run (grok epoch ~24,750, see l2_norm_4head_40k_trough_signal.md)
   - Single-run predictions would still be unreliable without more seeds

2. L2 NORM VARIABILITY:
   - Initial L2 norm: 115.6 to 116.4 (consistent)
   - Final L2 norm: 38.9 to 65.2 (variable)
   - Total decay: 50.5 to 77.3 (43.6% to 66.4% of initial)
   - L2 norm alone cannot predict grokking onset via a fixed threshold

3. DROPOUT GAP SIGNAL (CONCERNING, unaffected by the epoch-axis bug):
   - All runs show NEGATIVE dropout gap (-0.97 to -0.98)
   - This means dropout INCREASES accuracy after training
   - Expected: dropout should DECREASE accuracy (stress test)
   - This indicates either a bug in the dropout gap calculation, or the
     model has reached perfect/near-perfect accuracy (ceiling effect)
   - Dropout Gap needs care before being read as a predictor signal

4. LOSS DECAY:
   - All runs reach extremely low final loss (< 0.002)
   - Indicates perfect or near-perfect training fit
   - Loss curve does not clearly predict grokking on its own

5. BONUS -- TROUGH-TO-GROK LEAD TIME (not present in the original report):
   - Run 1: trough at epoch 17,920 (L2=49.9), grok at 18,893, lead = 973 epochs (5.2% of grok epoch)
   - Run 2: trough at epoch 22,524 (L2=48.0), grok at 29,982, lead = 7,458 epochs (24.9% of grok epoch)
   - Run 3: trough at epoch 22,464 (L2=47.9), grok at 24,572, lead = 2,108 epochs (8.6% of grok epoch)
   - Lead is positive (genuinely leading) in every run, but not tight or
     consistent as a percentage of run length -- this alone does not yet
     pass the project's 3-criteria validation protocol

6. RECOMMENDATIONS:
   x L2 Norm fixed-threshold rule: not a reliable standalone predictor
   x Dropout Gap: needs its calculation checked before use
   -> Trough-to-grok lead time is worth pursuing further with more seeds
   -> Per the project's evaluation order, Dropout comes right after L2
      Norm; this report already exercises Dropout Gap above, so Spectral
      is the natural predictor to take up next
   -> Ensemble multiple runs for any prediction attempt

7. EXPERIMENTAL RIGOUR:
   - Use a minimum of 3-5 runs before drawing conclusions
   - Report mean, standard deviation, min and max of metrics
   - Grokking is inherently stochastic; always sanity-check a new grok
     epoch reading against previously established runs before trusting it
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FINAL VERDICT

Can L2 Norm be used as a grokking predictor?

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ANSWER: NO.

L2 Norm was evaluated on BOTH transformer variants used in this thesis --
the single-head baseline and the four-head (Nanda-style) variant -- and in
BOTH cases it failed to produce a reliable, live-usable grokking predictor.
This is a genuine, project-wide negative result for Predictor 1 of 9.

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1. SINGLE-HEAD TRANSFORMER (baseline model)
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Five detection strategies were tried on three independent single-head
runs (grok epochs 5739 / 4806 / 3760):
   1. Raw rate-of-decline threshold        -- abandoned, signal inverted
   2. Second-derivative inflection          -- abandoned, dominated by noise
   3. Double-smoothed inflection             -- abandoned, superseded by (4)
   4. Fast/Slow MA crossover (50/200)        -- abandoned, fired 3000-5000+
                                                epochs too early every run
   5. MA-of-MA zero-crossing                 -- fired correctly on 2 of 3 runs,
                                                but fired 1884 epochs AFTER
                                                grokking on Run 3: disqualified
A sixth candidate, peak-of-difference, passed both the 'always leads' and
'tight, consistent gap' criteria (gap shrank from 1.05% to 0.18% of the run)
but is NON-CAUSAL -- it needs the full future curve to locate its peak, so
it cannot fire during live training.
VERDICT (single-head): formally closed as a NEGATIVE RESULT.

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2. FOUR-HEAD TRANSFORMER (this report's model)
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The raw L2 Norm curve itself is not usable here either: Final L2 Norm
ranges from 38.9 to 65.2 and total decay from 43.6% to 66.4% across three
runs with identical architecture and hyperparameters -- there is no fixed
threshold or shape that reliably marks grokking.
A trough-based reformulation (wait for the L2 Norm minimum, then confirm
a sustained rise) was checked against the project's own 3-criteria test,
using the three corrected runs above:
   Run 1: trough epoch 17,920, grok epoch 18,893, lead 5.2% of grok epoch
   Run 2: trough epoch 22,524, grok epoch 29,982, lead 24.9% of grok epoch
   Run 3: trough epoch 22,464, grok epoch 24,572, lead 8.6% of grok epoch
   Criterion 1 (always leads, never postdictive):  PASSES (all 3 runs)
   Criterion 2 (tight, consistent gap as % of run): FAILS (5.2%-24.9%,
                                                     nearly a 5x spread)
   Criterion 3 (clearly above noise floor):        not yet formally tested
VERDICT (four-head): promising direction, but NOT YET A VALIDATED
predictor -- cannot currently be used live.

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OVERALL CONCLUSION
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L2 Norm was tried on BOTH the single-head and the four-head transformer.
In BOTH cases, no version of it (raw curve, fixed threshold, MA crossover,
MA-of-MA zero-crossing, or the trough reformulation) cleared this project's
own validation bar for a usable, causal, live grokking predictor. L2 Norm
is CLOSED as a predictor for this benchmark. The trough signal remains the
one open thread worth more seeds if pursued later; otherwise, per the
project's evaluation order, the next predictor to take up is Spectral.
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